

Apply autumn reasoning

Use the scaffold, show each step and ask for an adult check when marked.

Name: _____ Date: _____

1. Read the model: A school buys 32 packs of 145 books and gives 18 books to each of 240 pupils. How many books remain? Copy or complete the first step.

2. A journey starts at -12 m, rises 37 m, then drops 18 m. Final position?

3. A budget of £20,000 pays for 48 items at £375 each. How much remains?

4. Miro says: Miro completes correct calculations but answers a different question from the one asked. Is this correct? Explain.

Teacher answers and checking prompts

1. Read the model: A school buys 32 packs of 145 books and gives 18 books to each of 240 pupils. How many books remain?
Copy or complete the first step.
Answer 320 books.
2. A journey starts at -12 m, rises 37 m, then drops 18 m. Final position?
Answer 7 m.
3. A budget of £20,000 pays for 48 items at £375 each. How much remains?
Answer £2,000.
4. Miro says: Miro completes correct calculations but answers a different question from the one asked. Is this correct? Explain.
Answer Return to the final question, label the answer and test whether it is reasonable in context.

Apply autumn reasoning

Work independently. Show a complete method and check at least one answer.

Name: _____ Date: _____

1. A school buys 32 packs of 145 books and gives 18 books to each of 240 pupils. How many books remain?

6. Check this result using an inverse, estimate or known fact: A budget of £20,000 pays for 48 items at £375 each. How much remains?

2. A journey starts at -12 m, rises 37 m, then drops 18 m. Final position?

7. Write a mixed problem using rounding, multiplication and subtraction, then solve it.

3. A budget of £20,000 pays for 48 items at £375 each. How much remains?

8. Identify and correct this misconception: Miro completes correct calculations but answers a different question from the one asked.

4. Solve this independently and show every stage: A school buys 32 packs of 145 books and gives 18 books to each of 240 pupils. How many books remain?

9. Write a complete sentence explaining how you checked one answer.

5. Use a second method or representation for: A journey starts at -12 m, rises 37 m, then drops 18 m. Final position?

10. Write and solve a similar question to: A budget of £20,000 pays for 48 items at £375 each. How much remains?

Teacher answers and checking prompts

1. A school buys 32 packs of 145 books and gives 18 books to each of 240 pupils. How many books remain?
Answer 320 books.
2. A journey starts at -12 m, rises 37 m, then drops 18 m. Final position?
Answer 7 m.
3. A budget of £20,000 pays for 48 items at £375 each. How much remains?
Answer £2,000.
4. Solve this independently and show every stage: A school buys 32 packs of 145 books and gives 18 books to each of 240 pupils. How many books remain?
Answer 320 books.
5. Use a second method or representation for: A journey starts at -12 m, rises 37 m, then drops 18 m. Final position?
Answer Expected result: 7 m. Method or representation may vary.
6. Check this result using an inverse, estimate or known fact: A budget of £20,000 pays for 48 items at £375 each. How much remains?
Answer Expected result: £2,000. A valid check must be shown.
7. Write a mixed problem using rounding, multiplication and subtraction, then solve it.
Answer Answers vary. The calculation and rounding decision must be justified.
8. Identify and correct this misconception: Miro completes correct calculations but answers a different question from the one asked.
Answer Return to the final question, label the answer and test whether it is reasonable in context.
9. Write a complete sentence explaining how you checked one answer.
Answer Answer should name an inverse, estimate, boundary, known fact or equivalent representation.
10. Write and solve a similar question to: A budget of £20,000 pays for 48 items at £375 each. How much remains?
Answer Answers vary. The new question must use the same mathematical structure and include a correct solution.

Apply autumn reasoning

Prove, compare or generalise. Write enough reasoning for another pupil to follow.

Name: _____ Date: _____

1. Write a mixed problem using rounding, multiplication and subtraction, then solve it.

6. Create a new example that tests the same idea as: A budget of £20,000 pays for 48 items at £375 each. How much remains?

2. Prove the answer to this question, rather than only calculating it: A school buys 32 packs of 145 books and gives 18 books to each of 240 pupils. How many books remain?

7. Find a second method or representation. Compare its efficiency with your first method.

3. Solve this in two different ways and compare them: A journey starts at -12 m, rises 37 m, then drops 18 m. Final position?

8. Write one always, sometimes or never statement about today's learning and justify it.

4. Work backwards from the answer to reconstruct the question: £2,000.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.

5. Explain why this reasoning fails: Miro completes correct calculations but answers a different question from the one asked.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.

Teacher answers and checking prompts

1. Write a mixed problem using rounding, multiplication and subtraction, then solve it.
Answer Answers vary. The calculation and rounding decision must be justified.
2. Prove the answer to this question, rather than only calculating it: A school buys 32 packs of 145 books and gives 18 books to each of 240 pupils. How many books remain?
Answer Expected result: 320 books. Proof or justification must match the lesson structure.
3. Solve this in two different ways and compare them: A journey starts at -12 m, rises 37 m, then drops 18 m. Final position?
Answer Expected result: 7 m. Two valid methods and a comparison are required.
4. Work backwards from the answer to reconstruct the question: £2,000.
Answer One valid reconstruction is: A budget of £20,000 pays for 48 items at £375 each. How much remains? Other equivalent questions are acceptable.
5. Explain why this reasoning fails: Miro completes correct calculations but answers a different question from the one asked.
Answer Return to the final question, label the answer and test whether it is reasonable in context.
6. Create a new example that tests the same idea as: A budget of £20,000 pays for 48 items at £375 each. How much remains?
Answer Answers vary. The example and solution must preserve the mathematical structure.
7. Find a second method or representation. Compare its efficiency with your first method.
Answer Answers vary. Comparison should refer to accuracy, number structure and clarity.
8. Write one always, sometimes or never statement about today's learning and justify it.
Answer Answers vary. A valid example and counterexample should support the classification.
9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.
Answer Answers vary. The error must be mathematically relevant and the correction must be precise.
10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.
Answer Answers vary. The explanation must distinguish the mathematical structure from the changed value.